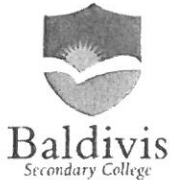


ANSWERS

Name:		Date: _____
Class:		



Baldivis
Secondary College

Year 9 Mathematics 2018

Investigation

Trigonometry

Validation

You will need: Collaborative component, calculator, ruler

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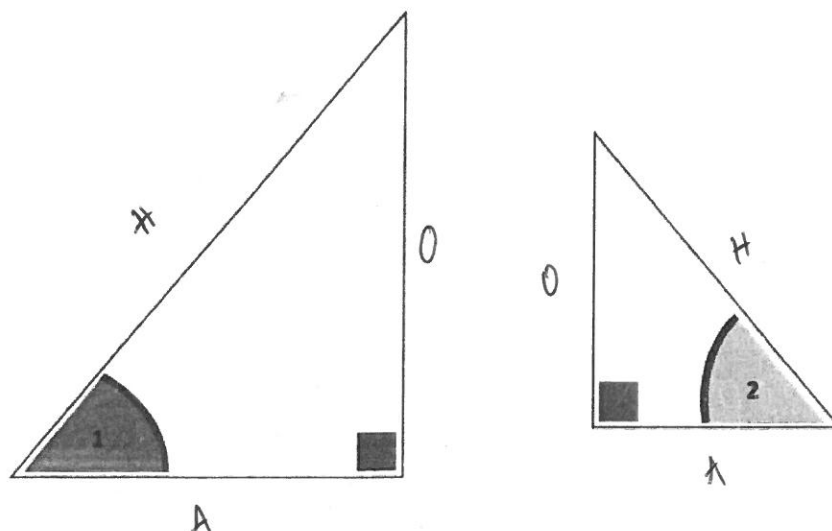
35 minutes

Question 1. Complete the table below using your COLLABORATIVE component.
(complete the 'ratios' column as a decimal value). [4 marks]

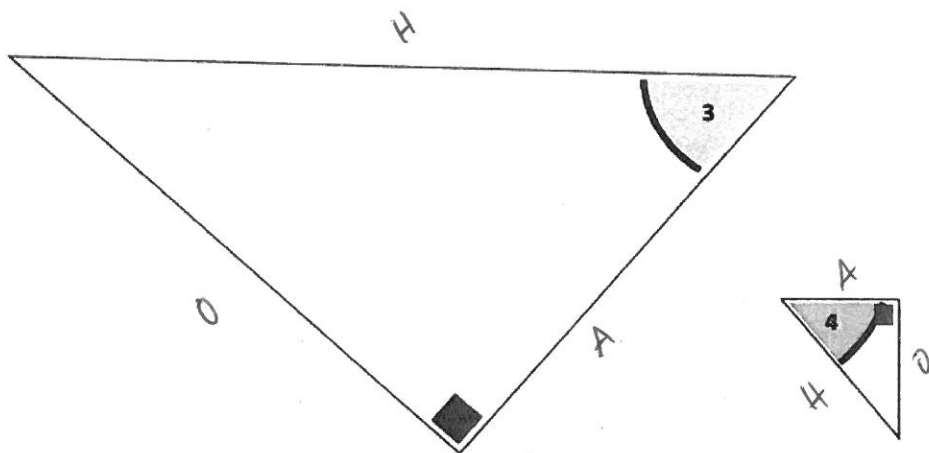
Triangle	Angle	OPP	ADJ	HYP	RATIOS		
					OPP ÷ HYP	ADJ ÷ HYP	OPP ÷ ADJ
2							
4							
6							
8							
				MEAN			

check for similarity in Ratios section

Question 2. Use the 4 triangles below to measure the lengths of each side and complete the table (complete the 'ratios' column as a decimal value). [5 marks]



& possibly check on calc



Triangle	Angle	OPP	ADJ	HYP	RATIOS			
					OPP ÷ HYP	ADJ ÷ HYP	OPP ÷ ADJ	
1	50°	62	52	81	0.77	0.64	1.19	✓
2	50°	39	33	51	0.76	0.65	1.18	✓
3	50°	79	66	104	0.76	0.63	1.20	✓
4	50°	18	15	24	0.75	0.63	1.20	✓
MEAN					0.76	0.64	1.19	

Question 3. Use a calculator to find;

[3 marks]

a) $\sin 50^\circ = 0.766$

b) $\cos 50^\circ = 0.642$

c) $\tan 50^\circ = 0.192$

Question 4. Comment on how the values relate between question 2 and 3? [1 mark]

- $\approx$ the same
- approximately the same ✓
- very similar

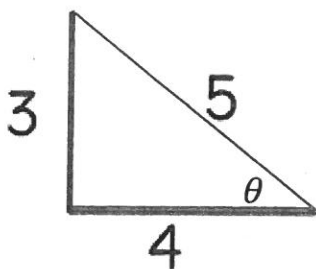
Question 5. Explain the relationship you've noticed in question 4, in terms of ratios for each angle size. [4 marks]

[You can use a diagram to explain and include words such as; **ratio**, **similar**, **angles**, **sides**, **hypotenuse**, **adjacent** and **opposite**]

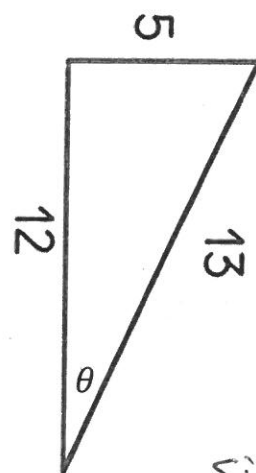
- diagram ✓ - 2 x triangles w same angle size [sides in proportion]
- similar triangles = similar ratios $\left[\begin{array}{l} \sin \\ \cos \\ \tan \end{array} \right]$
- triangles w same angle size will have hypotenuse, adjacent & opposite sides that are in proportion to each other

*** any other reasonable comment

Question 6. Given the two Pythagorean triads, state the sine ratio for the indicated angle. [2 marks]



$$\sin \theta = \frac{3}{5}$$



$$\sin \theta = \frac{5}{13}$$

Question 7. Make a comment on the two triangles above in relation to similarity.

- triangles are NOT similar ✓ [2 marks]
- sides [OPP / HYP / ADJ] are NOT in proportion

